

Combining Drift-Diffusion and Equivalent-Circuit Models for Efficient 3D Tandem Solar Cell Simulations

Andreas Fell¹, Oliver Schultz-Wittmann, Christoph Messmer², Martin C. Schubert, and Stefan W. Glunz³

Abstract—For upscaling silicon based tandem solar cells from small laboratory sizes to full size formats compatible with industrial production, two-dimensional (2-D) and 3-D effects like metal grid layout, perimeter design, and lateral inhomogeneities gain importance for tandem cell development. For understanding and quantifying such effects, 3-D tandem modeling is helpful, but the capabilities of existing solar cell simulation tools is limited in this respect. In this article, we describe a numerically efficient 3-D tandem modeling approach implemented in the solar cell simulation software Quokka3. It combines a 1-D equivalent-circuit (EQC) model of the top cell within the front side's boundary condition with either the quasi-neutral 3-D drift-diffusion model or an EQC model for the bottom cell's bulk carrier transport. This way the addition of a top cell to a single-junction silicon bottom cell model in Quokka3 adds little effort in terms of computational time and is thus compatible with large-area 3-D simulations up to full cell geometries. We showcase the usefulness of this approach by investigating various perimeter designs of small-area silicon-perovskite cells.

Index Terms—Edge losses, edge recombination, full-cell, modeling, passivated emitter and rear cell (PERC), quokka, silicon, simulation, solar cell.

I. INTRODUCTION

SILICON based tandem solar cells are currently experiencing high R&D efforts towards industrialization due to their proven efficiency potential substantially above the mainstream

silicon single junction cell technology. The most focused-on top cell option is a perovskite solar cell [1], [2], but there are also other options like in particular III-V solar cells [3]. The challenges to transfer the outstanding laboratory cell efficiencies into a marketable module are about long-term stability and upscaling cell sizes from the mostly tiny laboratory cells to full-size wafer formats [4]. For the latter, some important design considerations comprise: identifying the optimal metallization grid layout, which interplays with the lateral conductivity of functional films and minimizing perimeter losses, which is more complex compared to single-junction cells due to more functional material layers. Additionally, due to the higher number of functional layers and deposition steps, including, e.g., an organic perovskite absorber, layer nonuniformities leading to lateral inhomogeneities of cell characteristics are more pronounced compared to single junction silicon solar technology [5]. Spatially resolved characterization and quantification of the adverse impact on global cell characteristics of such lateral inhomogeneities is therefore supposed to be of particular importance for future tandem cell development, for which task 3-D tandem modeling is essential.

3-D electrical solar cell simulation is nowadays well established to describe the mentioned effects for single-junction silicon solar cells and has become an integral part for their further development. Many of the popular used solar cell simulation tools however either do not support multi-junction solar cells, or are restricted to a (quasi-)1-D representation of the solar cell and therefore not able to simulate 3-D tandem physics. Therefore the vast majority of published tandem solar cell simulation works deal with investigating and optimizing the (quasi-)1-D tandem physics [6], [7], [8], or focus on the bottom cell only [9], [10]. Investigation of tandem metallization grid layout is usually performed by, often analytical, series resistance modeling, strongly simplifying the electrical tandem physics into a constant current source, see, e.g., [11], [12].

The main reason for the lack of large-area 3-D tandem cell simulation capabilities is that most electrical solar cell simulation tools are based on solving the drift-diffusion (DD) equations within the involved materials [13], [14], [15], [16], [17], [18], [19]. However, fully solving the DD equations for a large-area tandem solar cell would require an extremely large numerical mesh and computation time, rendering this approach practically prohibitive beyond small unit cell domains. An alternative modeling approach is to represent the subcells as an equivalent circuit

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model (e.g., two series-connected two diode models), which comes at the cost of greatly reduced physical detail. On the other hand, the simplicity results in low numerical and user effort and is therefore a popular approach for tandem/multijunction cell modeling. When laterally connecting many of such EQC models, a so-called distributed diode model is achieved. Such a model has been established and used to investigate metal grid related effects for multijunction cells [20], [21]. However, to the knowledge of the authors, there exists no publicly available solar cell simulation tool yet implementing a distributed diode model specifically for tandem solar cells.

In this article, we address this lack of simulation capabilities by introducing an extension to the “skin concept” implemented in the solar cell simulation tool Quokka3 [15]. In this concept, the DD equations are solved in 3-D, but only within the quasi-neutral bulk, which typically represents the main part of the silicon wafer. All effects taking place close to the surfaces, i.e., within the “skins,” are lumped into few effective properties forming a boundary condition. This has been shown to be very successful in reducing the required numerical mesh size and computation time, being compatible with large-area 3-D solar cell simulations up to full wafer sizes. We describe how this skin boundary condition is extended to support an equivalent circuit model representing a top cell within a tandem cell.

Additionally, an EQC model type for the quasi-neutral bulk is implemented as an option. This offers the user of Quokka3 the flexibility to simulate a 3-D tandem cell either with a coupled 3-D DD bulk and EQC top cell model, or with a simpler distributed diode model.

We first describe the details of the model as implemented in Quokka3. Then we validate the model by comparisons against 2-D Sentaurus TCAD [16] simulations. Lastly, the usefulness of the new capabilities is showcased by an investigation of perimeter losses for laboratory tandem solar cells developed at Fraunhofer ISE.

II. MODEL DESCRIPTION

In the skin concept implemented in Quokka3 the solar cell is split into 3 essential parts: the quasi-neutral bulk, which commonly represents the main part of the silicon wafer; the laterally conductive skin, which comprises the near-surface region of the bulk material and the functional layers on top; and) the metal-lization grid which contacts the skin and conducts the current to the terminals. In the bulk, the quasi-neutral form of the DD equations are solved in 3-D in steady-state, closed via the lumped electrical properties of the skins as laterally conductive boundary conditions, which was introduced as the “conductive boundary model” [22]. The main lumped electrical properties of a skin are the sheet resistance R_{sheet} and the recombination property $J_{0\text{skin}}$. These two properties are experimentally well accessible, which rendered this modeling approach very useful and popular next to the conventional way of solving DD also within near-surface region, i.e., not treating the skin as a separate entity. By omitting the numerical mesh within the skin for the 3-D solver, and by being able to use the quasi-neutral simplification of the DD equations, the implementation in Quokka3 is numerically highly efficient

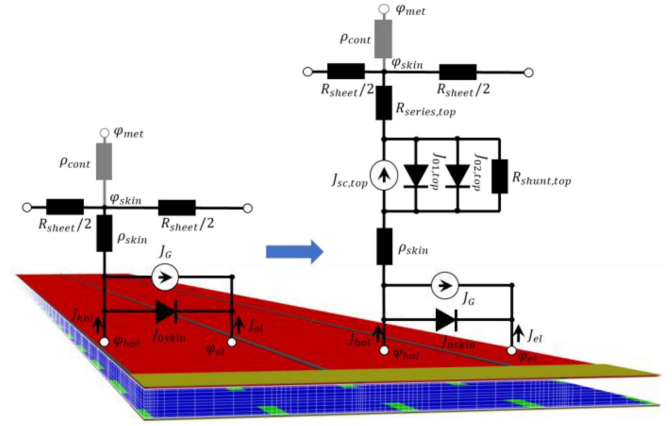


Fig. 1. Extension of the lumped skin EQC boundary condition to include a top cell JV-curve via a two-diode model; exemplary 3D PERC cell solution domain where the shown EQC boundary conditions are applied to each lateral mesh element, connected laterally via the sheet resistances.

and copes with large-area 3-D domains up to full-size solar cells.

The skin concept also enables a so called “multiscale” approach [23]. Here a skin is not defined by its lumped properties, but rather by its detailed properties like e.g., doping profile and surface recombination velocities. First, DD is solved in 1-D for each detailed skin, and from the results the lumped properties are automatically provided as a boundary condition to the subsequent 3-D bulk solver. These 1-D detailed skin simulations add little computational effort, meaning that the multiscale approach enables large-area 3-D simulation accounting for the detailed properties of the skin, with results equivalent to a full 3-D DD simulation with orders of magnitude higher computational demand. The results are arguably accurate for most wafer-based silicon solar cell simulation tasks, due to the skins being typically much thinner than other cell dimensions causing the imposed assumptions to be valid [15].

In the following, the model extensions are described for the case of a top cell within the front skin, representing, e.g., the perovskite top cell within a perovskite-silicon tandem cell. The approach however can well be generalized to a subcell being contained in a skin on any side of the device and is equally applicable for other top cell technologies.

Notably, the 3-D implementation in Quokka3 so far only solves for steady-state conditions, meaning that it cannot model transient effects, like top cell hysteresis effects.

A. Equivalent Circuit Model of Top Cell Skin

The mentioned lumped skin boundary condition can already be interpreted as an equivalent circuit model with the diode $J_{0\text{skin}}$, a current source J_G representing the collected generation current within the near-surface region, a vertical resistance ρ_{skin} representing, e.g., the resistance over a thin oxide within a passivating contact, the sheet resistance R_{sheet} , and a contact resistance ρ_{cont} to the metal layer (if present at this coordinate), see Fig. 1. The extension to support a top cell is realized by

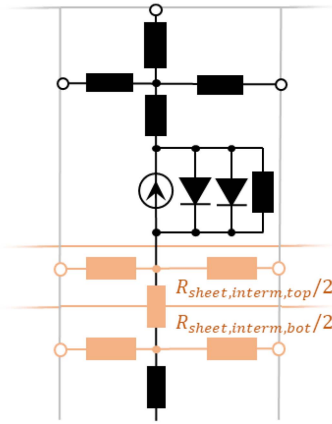


Fig. 2. Top cell skin boundary condition indicating a possible enhancement in orange to account for lateral conductance in the intermediate layers by adding two numerical mesh layers; the lower sheet resistance can also include the respective carrier type's bulk transport in the bottom cell within a distributed diode model.

adding an active JV -curve in series, which can well be parameterized via a two-diode model in most cases.

It must be noted that, in the current implementation in Quokka3, the skin layer still is a single laterally conducting boundary, i.e., is represented by a single skin potential φ_{skin} at each lateral coordinate. Within the numerical mesh there exist only φ_{skin} and the bulk-side electron and hole potentials φ_{el} and φ_{hol} from which the electron and hole current densities J_{el} and J_{hol} are calculated semi-analytically in each numerical solver iteration. Consequently, the sheet resistance represents lateral conduction only on top of the active part of the top cell, which is usually the front transparent conductive oxide (TCO) layer. Intermediate lateral conduction between the bottom and top cell is neglected in this implementation. This is a valid simplification for typical tandem cells, as lateral conduction between the subcells is no fundamental requirement for high cell performance, see also Section III. However, it may become inaccurate in cases of strong lateral variations, and in particular is insufficient to describe concepts with local contacts between the subcells, see, e.g., [10]. To overcome this limitation, two intermediate conducting layers need to be added to the numerical mesh between the subcells to represent lateral conduction in the top layers of the bottom cell, and in the bottom layers of the top cell, respectively (see Fig. 2).

It becomes obvious that in this lumped skin tandem approach the top cell electro-optical characteristics need to be known as inputs rather than being a simulation result. This shifts the applicability of this modeling approach away from optimizing the top cell layers to other tasks like metal grid or perimeter design investigations, bottom cell optimization, or investigation of lateral inhomogeneities. It is however noted that in principle the multiscale concept is also applicable to a tandem cell. Meaning that the equivalent circuit model parameters can also be populated from 1-D detailed simulation results and subsequently coupled to the 3-D bulk solver as boundary conditions. This can be done manually by parameterizing the JV -curves simulated with other software, and is planned to be automated in a future

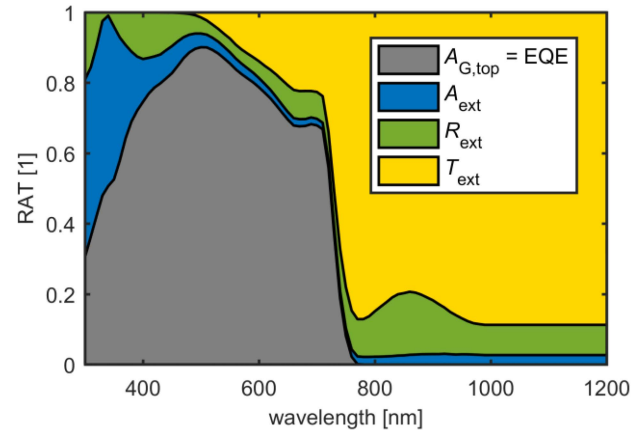


Fig. 3. Exemplary spectrally resolved reflection and absorption data of a perovskite top cell derived from detailed optical simulations, suitable as inputs for Quokka3's " T_{ext} -Z" model.

Quokka3 update using the existing 1-D solver suitable, e.g., for perovskite solar cells [24].

B. Optical Top Cell Modeling

Regarding the optical modeling, the presented approach requires the knowledge or determination of the top cell's short-circuit current density, $J_{\text{sc,top}}$. One option is to directly set a value for $J_{\text{sc,top}}$, and additionally the generation (profile) in the silicon bottom cell, which may be determined by separate optical modeling, e.g., ray-tracing, or deduced from measurements. The more detailed option is to input the spectrally resolved optical characteristics of the top cell. Those are essentially the spectrally resolved active absorptance $A_{G,\text{top}}$ (which equals in good approximation the top cell external quantum efficiency EQE_{top}), the external reflection R_{ext} and the parasitic absorption during the first pass of light in all the top cell's material layers A_{ext} . This sums up to the external transmission T_{ext} , which defines the fraction of light entering the bottom cell (see Fig. 3)

$$T_{\text{ext}}(\lambda) = 1 - R_{\text{ext}}(\lambda) - A_{\text{ext}}(\lambda) - A_{G,\text{top}}(\lambda).$$

$J_{\text{sc,top}}$ is then found by integrating $A_{G,\text{top}}$ over the incident spectrum. Notably, this spectrally resolved optical definition of the top cell is compatible with the " T_{ext} -Z" model implemented in Quokka3 [25], which allows spectrally resolved electro-optical modeling, e.g., for quantum efficiency simulations.

C. Equivalent Circuit Model for Bottom Cell

Here, another optional simplification to the bulk current transport solver is introduced, namely replacing the 3-D DD solver on a numerical mesh by an EQC model, see Fig. 4. This eliminates the requirement for a numerical discretization in the depth-coordinate and creates effectively a distributed diode model of the solar cell. Lateral conduction in the bulk for both majority and minority carriers, and also a resistance between the bulk and the skin (e.g., over a thin oxide), is considered analytically in Quokka3 by calculating a single effective value

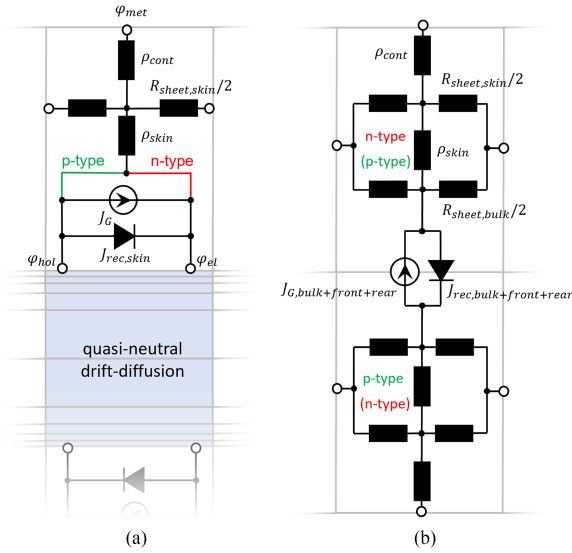


Fig. 4. Two bulk model types in Quokka3 (a) 3-D quasi-neutral DD in the discretized bulk with lumped skin equivalent circuit boundary conditions. (b) EQC modeling with a current source and diode representing total generation and recombination in the bulk and skins, resulting in a distributed diode solar cell model.

for the sheet resistance for the front and rear skin. Notably, the input parameters, i.e., the electrical and optical properties of the bulk and skins, are identical irrespective of the bulk model type. This means that within Quokka3, switching between the 3-D DD and distributed diode model is possible by a single option in the settings.

The benefits of the EQC bulk model type are greatly reduced computational demand and additionally increased numerical robustness, e.g., a numerical issue when solving for small applied voltages for an illuminated solar cell, in particular short-circuit conditions, is alleviated with the new bulk model type. However, the drawback is a lack of physical detail, mainly the assumption of a depth-independent Fermi-level split. This results in the inability to resolve current crowding effects at local contacts, nonideal current collection, diffusion-limited recombination, etc. Also, the implementation in Quokka3 strictly allows only a single skin polarity per side, making it incompatible with, e.g., interdigitated back contact solar cells. It thus depends on the concrete solar cell design and simulation task whether the EQC bulk model type is suitable and advantageous. It is in particular useful for metal-grid investigations on large solution domains but should be avoided if speed and robustness are no practical limitation.

For tandem cell simulations with an EQC top cell model the EQC bulk model type appears the natural match and is thus the recommended option for high numerical robustness and speed. However, it can be useful or even necessary to use the 3-D DD model type in some cases, e.g., if a detailed optimization of the bottom cell is the task. Furthermore, the limitation of a single conducting layer per side in the current implementation eliminates lateral conduction in the bulk for the carrier type collected at the front side, i.e., for a front / rear emitter bottom cell, lateral conductance via the bulk's minority/majority carriers

and via the bottom cell's front conducting layers is missing. This simplification can be alleviated by adding intermediate conducting mesh lasers (see Fig. 2), or in the current implementation by using the DD bulk model type to still account for lateral bulk transport effects (see Section III).

III. MODEL VALIDATION

In order to validate the proposed model and its implementation, we compare tandem simulation results of Quokka3 with Sentaurus TCAD [16], in particular to test the aforementioned simplifications regarding the missing intermediate lateral conductance.

In Sentaurus, we perform a fully detailed optoelectrical simulation of a perovskite-silicon tandem cell featuring a 250 μm thick 1 Ωcm p-type cSi bottom cell with an ideal nonrecombining full-area rear contact, 1 ms SRH bulk lifetime, and an ideal nonrecombining *n*-type poly-Si front emitter. The top cell consists of a state-of-the-art planar *p-i-n* perovskite layer stack as used in [26] featuring a 20 nm intermediate indium tin oxide (ITO) layer as recombination TCO, a 1 nm thin 2PACz layer as hole transport layer, a 450 nm thick perovskite layer, and a C60/SnO₂ layer stack as ETL. The front TCO features lateral conductance of 150 $\Omega/\square$. The cell is solved optically via ray-tracing and transfer-matrix-method according to several prior publications [10], [11]. The electrical simulation of the full tandem cell solves the semiconductor equations within the whole 2-D element including the complex thin-film layer stack. For the comparison with Quokka3, *JV* hysteresis due to mobile ions within the perovskite was neglected.

We first perform a quasi-1-D simulation of the tandem cell in Sentaurus TCAD. The top cell's light and dark *JV*-curves are extracted and subsequently fitted by a two-diode model. Notably the two-diode parameters are significantly different between the light and dark case, indicating that the two-diode model should be considered a mathematical parameterization rather than an accurate physical model of the perovskite top cell. We do not want to speculate at this point whether this is something typical for perovskite cells, or particular for the chosen theoretical properties, but it has implications to the following 2-D simulations, see below.

In Quokka3 we set the optical generation currents in the sub cell same as in the Sentaurus simulations, assume the same bottom cell bulk properties and zero surface recombination, and apply the two-diode model parameters to the top cell. A close match of the *JV*-curves between Quokka3 and Sentaurus is observed, with the light *JV* parameters deviating less than 0.1%, evidencing that the basic sub cell series connection is correctly modelled in Quokka3.

Next we define the same 2-D perimeter domain in both tools, see Fig. 5. In the perimeter region no illumination is present, and optionally an isolation between the subcells is defined. This introduces strong lateral variations in order to reveal any potential influence of laterally conductive layers between the subcells. The isolation is achieved in Quokka3 by setting a very high vertical skin resistivity, and in Sentaurus by adding a thin oxide layer between the poly-Si and TCO layer without any

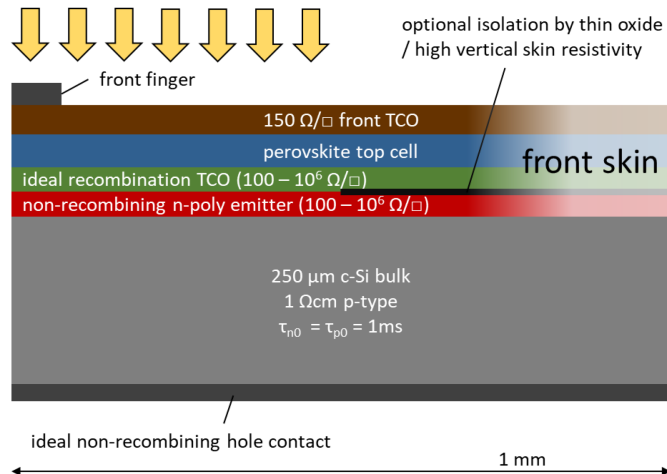


Fig. 5. Sketch of the half-illuminated 2-D perimeter domain created both in Sentaurus and Quokka3; in Quokka3 the lateral conduction of the intermediate layers (n-poly emitter and recombination TCO) is not accounted for, but is varied in Sentaurus.

tunneling model activated. Notably, in the dark perimeter we apply the dark two-diode model parameters, which was found to be critical to achieve a good match of the 2-D simulation results.

As expected, due to the large mesh required in Sentaurus for discretization of the thin layers and the detailed physics solved additionally for the top cell's layer stack, the simulation time is much higher compared to Quokka3. A single *JV*-curve simulation takes around half an hour on 6 CPU cores, while with Quokka3 on a single CPU it takes around one second for the quasi-neutral and another order of magnitude less for the EQC bulk model type, respectively.

In Sentaurus, we further vary the sheet resistance of the poly-Si emitter and recombination TCO layers by manually adjusting their mobility to achieve very high ($10^6 \Omega/\square$), high ($10^4 \Omega/\square$), and low ($100 \Omega/\square$) sheet resistance values.

We then calculate an edge-length specific current density loss into the perimeter by calculating the current density difference of the 2-D simulations to the 1-D reference simulation and multiplying by the width of the solution domain of 1 mm. This is evaluated both at the open-circuit and maximum power point voltage of the 1-D simulations.

The results in Fig. 6 show a reasonable agreement between Quokka3 and Sentaurus, evidencing that Quokka3's skin concept is in general able to account for lateral current transport effects also for the tandem extension. In particular, for the default case of connected subcells, both the DD and EQC bulk model type accurately predict the dark perimeter losses. For connected subcells, no influence of the intermediate sheet resistances is observed in the Sentaurus simulations, evidencing its neglect in the Quokka3 model to be a non-detrimental simplification in this case.

For the case of isolated subcells, a reduced perimeter loss is observed both in Sentaurus and Quokka3. However, the EQC model type fails to predict a still substantial remaining perimeter loss, whereas the DD type is still able to correctly predict the

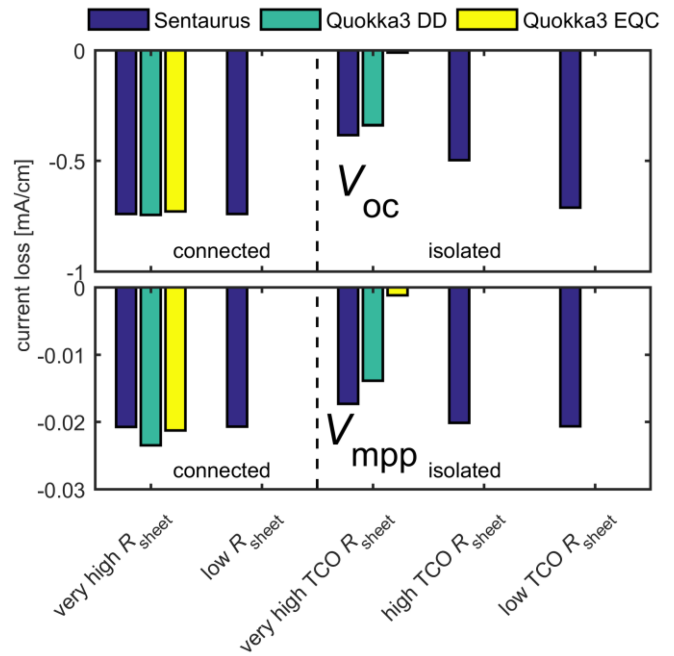


Fig. 6. Edge-length specific current density loss into the perimeter at open-circuit (V_{oc}) and maximum power point (V_{MPP}) voltage for various cases of intermediate sheet resistance values; comparing simulation results from Sentaurus and Quokka3 with the DD and EQC bulk model type; Quokka3 results are only comparable at very high intermediate sheet resistances.

perimeter loss for the case of high recombination TCO sheet resistance. This is due to the fact that in the DD case it is possible for both electrons and holes to be conducted laterally in the perimeter of the cSi bottom cell where they recombine. It is thereby evidenced that this remaining perimeter loss is attributable to the bottom cell. Due to the very low lateral conductance of the thin perovskite layer, a perimeter loss in the top cell requires lateral conduction both at the front (electrons) and rear (holes) of the bottom cell, and is thus only present for not very high recombination TCO sheet resistance. For low sheet resistance values, the top cell perimeter loss is thus activated, and the total perimeter loss approaches that of the connected case. This top cell perimeter loss for the isolated case cannot be replicated by Quokka3 due to the missing intermediate sheet resistance. It is expected that a future extension of the numerical model as indicated in Fig. 2 would result in correct perimeter losses in Quokka3 for all cases and for both bulk model types.

Not shown are the Sentaurus results with the variation of the *n*-poly sheet resistance. This has been observed to have a negligible impact due to the minority carrier transport in the c-Si bulk already providing a low sheet resistance for electrons.

Finally, we apply edge recombination in Sentaurus via defining a surface recombination velocity of 10^7 cm/s both at the edge of the perovskite and silicon layer. It is found that the resulting edge loss is strongly dominated by the recombination at the silicon edge (>1 order of magnitude higher than at the perovskite edge). This is expected due to the large difference in layer thickness and allows the neglect of perovskite top cell

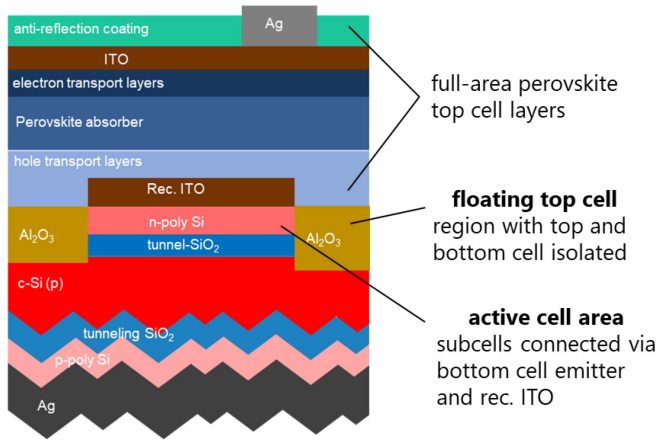


Fig. 7. Cross section sketch of the laboratory tandem cell design featuring a perimeter region with isolated top and bottom cell as an idea to suppress perimeter losses.

edge recombination when simulating edge recombination losses of tandem cells, as is the case in Quokka3.

IV. PERIMETER DESIGN OF 1X1CM² TANDEM CELLS

Most laboratory perovskite-silicon tandem cells are designed via using wafer-size silicon bottom cells and placing small perovskite top cells on it, which are measured via a shadow mask (“designated area” measurement). In particular for highest efficiency devices, the active cell size is around 1 cm x 1 cm or even smaller, to relax requirements on thin-film processing uniformity, and also to avoid a metal grid and thus shading losses within the active cell area.

One downside of this design is that the inactive bottom cell perimeter leads to carrier transport losses into the dark perimeter region. As the active cell area to perimeter ratio is rather low for these small sizes, these losses can be substantial, and it is thus a worthwhile effort to design the perimeter region in such way to minimize these losses. Because experimental investigation of various perimeter designs is of substantial effort, not the least due to often moderate reproducibility of laboratory solar cell batches, numerical tandem simulation can effectively support this task.

Here, we investigate one particular perimeter design idea, namely employing full-area top cell films, and structuring only the bottom cell emitter and recombination ITO layers to define the small active area. The idea is that the electrical isolation of the subcells within the perimeter region may minimize the perimeter losses, with a relatively straightforward manufacturing process flow, see Fig. 7.

We set up a quarter symmetry solution domain of such 1 cm x 1 cm tandem cells in Quokka3, where the illumination is restricted to the active cell area, as in the experimental designated area measurement, see Fig. 8. The required electro-optical inputs are roughly calibrated to the Fraunhofer ISE tandem cells achieving an efficiency of $\sim 25\%$ [26], see Table I for details on the input parameters. The only difference between the active and perimeter region besides the illumination is the definition of a “vertical skin resistance,” i.e., the isolation resistance, which is the resistance between the bottom cell bulk and the top-cell.

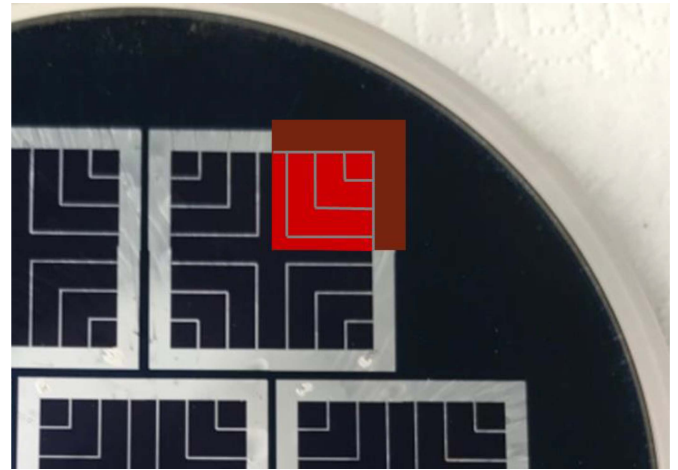


Fig. 8. Photo of multiple 1 cm x 1 cm tandem cells on a full-wafer, overlaid with the Quokka3 quarter-symmetry solution domain including a nonilluminated perimeter region.

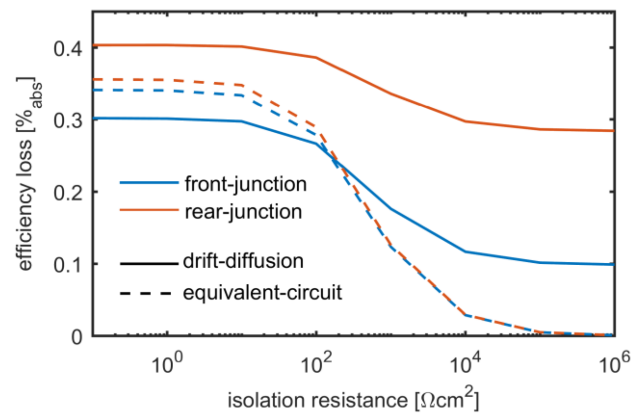


Fig. 9. Perimeter loss of 1 cm x 1 cm laboratory tandem cells as a function of the isolation resistance; with 2 mm perimeter width and variation between front- and rear-junction cell; using the EQC bulk model type results in significant inaccuracies compared to the quasi-neutral DD model type due to the current implementation neglecting lateral electron transport in the Si bottom cell.

For a very high isolation resistance the top cell in the perimeter region is then called “floating” top cell.

As a reference point, we simulate a device with zero perimeter width, which consequently results in zero perimeter loss. The efficiency loss with perimeter is then calculated by the difference to the reference efficiency.

To first identify a requirement on the isolation resistance, we vary its value over orders of magnitude for a fixed perimeter width of 2 mm. Also, the doping type of the bulk is varied between *p*-type and *n*-type, effectively varying between a front- and rear-junction bottom cell, which notably result in very similar absolute efficiencies. In Fig. 9, it can be seen the perimeter loss for the investigated cell design can be considerable, in particular for a rear junction device. This is to be expected, as the rear junction combined with the full-area rear metallization supports the minority carriers in the bottom cell to be transported effectively into the perimeter to recombine. Notably, the front vs. rear junction difference is large because of a very high emitter sheet resistance, and would be strongly reduced when applying lower emitter

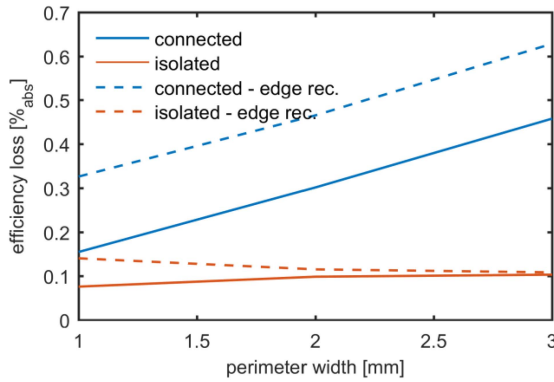


Fig. 10. Perimeter loss of 1 cm x 1 cm laboratory tandem cells with connected and isolated subcells in the perimeter region, and with and without worst-case edge recombination [23] (on the silicon bottom cell edge only).

sheet resistance values typical for single-junction cells, which is however not beneficial for a tandem cell. For the investigated front-junction bottom cell design, the larger part of the perimeter losses can indeed be suppressed via isolating the subcells. It requires a high-quality isolation resistance $> 1000 \Omega\text{cm}^2$, which however appears feasible with the intended aluminum-oxide layer and the omission of the recombination junction.

Also, we use this investigation to highlight some limitations of the EQC bulk model type. When enabling it instead of the quasi-neutral DD type, the Quokka3 simulations are 1-2 orders of magnitudes faster (seconds vs. minutes for a JV -curve) and give within $\sim 0.2\%$ the same results for the perimeter loss free reference cases, see Table II. However, the EQC type fails to accurately resolve the current transport in the perimeter, for the reasons discussed in Section III. Fig. 9 shows that the difference between front- and rear-junction cannot be reproduced, which is understandable as the EQC model is identical regardless of the position of the junction. Also, the still substantial remaining perimeter losses for very high isolation resistance are then missed out on, as then the effective sheet resistance for one of the carrier types becomes extremely high.

Next, we vary the perimeter width, which represents the (half) distance between neighboring active cells on the wafer, or to the actual cell edge, e.g., for cut-out cells. We perform the simulation with zero and very high ($10^6 \Omega\text{cm}^2$) isolation resistance. Also, we include the effect of edge recombination of the silicon bottom cell, assuming zero and “worst-case” edge recombination, the latter representing clean cut-edge with zero passivation effect [23]. We furthermore assume that edge recombination of the (floating) perovskite top cell is negligible, which is a reasonable assumption given its much smaller thickness (see also discussion in Section III). In Fig. 10, it can be seen that without vertical isolation, which is essentially a uniform tandem cell measured on a smaller area by shading of the perimeter, the perimeter losses scale quite linearly with perimeter width, and that edge recombination would induce a further substantial loss. When introducing the isolation, all edge losses are effectively suppressed, pretty much independent of the perimeter width within the practical range.

This proves that such an isolation by design is a very sensible idea, and one expects a residual perimeter loss of around only 0.1% .

V. CONCLUSION

We presented an extension of the “skin-concept” implemented in Quokka3 to support an EQC model of a subcell, which allows steady-state electro-optical 3-D tandem cell simulations with a particular focus on wafer-based c-Si bottom cells. As the extension is contained within the skin boundary condition to the 3-D bulk carrier transport solver, the numerical mesh and computational demand is not largely affected, enabling large-area 3-D solution domains as with single-junction cells.

The required inputs are the electro-optical characteristics of the subcell, namely the JV -curve, parameterized into a two-diode model, and the spectrally resolved EQE and transmission to the bottom cell. Similar as with the single-junction skin properties J_0 and R_{sheet} , these top cell inputs can be determined from cell-level measurements. This makes the approach agnostic to the top cell technology, and substantially simpler to apply compared to a full 3-D DD tandem model, which requires many hard-to-determine properties of the involved materials and interfaces. Notably, the inputs can also be populated from detailed 1-D DD simulations, resulting in a tandem multiscale model, which is planned to be implemented in Quokka3 in future as well.

Also, we presented a new bulk model type in Quokka3, by simplifying the 3-D DD model to an EQC model without a depth discretization. This increases numerical speed and robustness, at the cost of neglecting some 3-D transport related effects in the c-Si cell and can easily be switched by a single setting in Quokka3. If combined with a top cell skin, a quasi-3-D distributed diode model of a tandem cell is created. This appears to be the natural choice for tandem modeling when already using an EQC model for the top cell and is in fact the recommended choice for most simulation tasks. There are however situations for which the distributed diode tandem model in its current implementation can be an oversimplification, e.g., for strong lateral inhomogeneities or perimeter effects. We show this to be related mainly to accounting only for a single carrier type for the bottom cell’s lateral bulk carrier transport and to neglecting intermediate conducting layers. The former is alleviated by using the DD bulk model type. A proper alleviation for both bulk model types requires the addition of two intermediate conducting layers in the numerical mesh, which is planned to be implemented in the future.

While not directly applicable to optimize the top cell properties themselves, the proposed model can be useful for many tasks ranging from: understanding basic tandem effects with relatively simple input parameters; accurate and comprehensive metal grid design optimization; detailed bottom cell optimization; quantification of the impact of lateral inhomogeneities on global cell performance; and investigating cell measurement setups within the simulation, e.g., masked illumination or probe positioning.

As an application example we investigated small-area laboratory Si-perovskite tandem cells employed on a full-wafer c-Si bottom cell, where the idea was to suppress perimeter losses

by isolating the top and bottom cells in the perimeter region. By respective Quokka3 simulations including the isolated dark perimeter we could show that the idea is in principle promising to suppress the larger part of the perimeter losses. We also found that a front-junction bottom cell design with high emitter sheet resistance is beneficial in any case to suppress minority carrier flow into the perimeter. The application example also highlights the limits of the implemented distributed diode model to quantify such perimeter losses, which can be alleviated by including additional laterally conducting layers in between the subcells.

APPENDIX

TABLE I

MAIN ELECTRO-OPTICAL INPUT PARAMETERS USED IN QUOKKA3 TO APPROXIMATELY REPRESENT THE ~25% EFFICIENT PEROVSKITE-SILICON TANDEM CELLS MANUFACTURED AT FRAUNHOFER ISE [26]

optics via “ $T_{\text{ext}}\text{-}Z$ ” option	
$T_{\text{ext}}, A_{\text{ext}}, A_{G,\text{top}}$	from detailed optical simulation, see Fig. 3
bottom cell light trapping	Z-parameterization calibrated to detailed optical simulation results: $Z_0 = 10$; $Z_{\text{inf}} = 1$; $Z_p = 3.818$
silicon bottom cell electrical properties	
bulk doping	1.5 Ωcm p-type
bulk lifetime	intrinsic and 3ms SRH
front J_0	10 fA/cm ²
rear J_0	20 fA/cm ²
perovskite top cell EQC parameters	
J_{01}	$4 \cdot 10^{-21}$ A/cm ²
J_{02}	$5 \cdot 10^{-12}$ A/cm ²
R_{series}	2.5 Ωcm^2
R_{shunt}	10^{10} Ωcm^2
front R_{sheet}	150 Ω/sq

TABLE II

IV PARAMETERS OF THE SIMULATED TANDEM CELL WITHOUT PERIMETER LOSSES

bulk model type	V_{oc} [V]	J_{sc} [mA/cm ²]	FF [%]	η [%]
drift-diffusion	1.776	17.5	80.48	25.02
equivalent-circuit	1.776	17.5	80.63	25.06

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